

Conditional Polynomial Probabilistic Dimension from Stable ReLU Gates

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1 Theoretical Setup

1.1 Problem setting

Let $n \in \mathbb{N}_{\geq 1}$, let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, and put $\mathcal{X} = \{-1, +1\}^n$ and $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$. We write $\Delta(\mathcal{X})$ for the probability distributions on $\mathcal{X}$ and $[m] = \{1, \dots, m\}$. Fix the source tie label $s_0 = \text{sign}(0) \in \{-1, +1\}$ and define

$$\text{sign}_{s_0}(z) = \begin{cases} +1, & z > 0, \\ -1, & z < 0, \\ s_0, & z = 0. \end{cases}$$

For a map $\varphi : \mathcal{X} \rightarrow \mathbb{R}^d$, a coefficient $w \in \mathbb{R}^d$, $\mathcal{D} \in \Delta(\mathcal{X})$, and $h \in \mathcal{H}$, define the tie-resolved linear error

$$R_{\mathcal{D},h}(w, \varphi) := \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(\langle w, \varphi(x) \rangle) h(x) < 0].$$

For $\alpha \geq 0$, the probabilistic dimension complexity is

$$\text{dc}_\alpha(\mathcal{H}) := \min \left\{ d \in \mathbb{N}_0 : \begin{array}{l} \exists \text{ a law } \mathcal{P} \text{ over } \varphi : \mathcal{X} \rightarrow \mathbb{R}^d \quad \forall \mathcal{D} \in \Delta(\mathcal{X}) \quad \forall h \in \mathcal{H}, \\ \mathbb{E}_{\varphi \sim \mathcal{P}} \left[\inf_{w \in \mathbb{R}^d} R_{\mathcal{D},h}(w, \varphi) \right] \leq \alpha \end{array} \right\}. \quad (1)$$

Thus the feature-map law is selected before every later distribution and target, whereas the optimizing coefficient may depend on the realized map, distribution, and target. The fixed tie label makes the definition nonvacuous at a zero score.

Fix a depth $L \in \mathbb{N}_{\geq 1}$, positive integer widths $n_0 = n, n_1, \dots, n_{L-1}, n_L = 1$, and matrices $\theta_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$. The network has no biases and has parameter count

$$S := \sum_{\ell=1}^L n_\ell n_{\ell-1}, \quad \theta = (\theta_1, \dots, \theta_L) \in \mathbb{R}^S. \quad (2)$$

With $\sigma(a) = \max\{0, a\}$ coordinatewise, define

$$z_0(\theta, x) = x, \quad u_\ell(\theta, x) = \theta_\ell z_{\ell-1}(\theta, x), \quad z_\ell(\theta, x) = \sigma(u_\ell(\theta, x)) \quad (1 \leq \ell < L),$$

and $f_\theta(x) = \theta_L z_{L-1}(\theta, x)$. The coordinate $u_{\ell,j}$ denotes the j th entry of u_ℓ .

Fix the source protocol's ReLU-kink convention $\kappa \in [0, 1]$ and write

$$\rho_\kappa(a) = \begin{cases} 1, & a > 0, \\ 0, & a < 0, \\ \kappa, & a = 0. \end{cases}$$

The notation $\nabla_{\theta}^{(\kappa)}$ denotes the resulting single-valued back-propagation derivative. Both s_0 and κ are fixed protocol conventions, not choices made by the theorem.

Initialize all weights independently according to

$$(\theta_{\ell}^{(0)})_{jk} \sim \mathcal{N}(0, 1/n_{\ell-1}), \quad 1 \leq \ell \leq L. \quad (3)$$

For $\mathcal{D} \in \Delta(\mathcal{X})$ and $h^* \in \mathcal{H}$, draw $x^{(0)}, \dots, x^{(T-1)}$ independently from $\mathcal{D}$, independently of $\theta^{(0)}$. With $\ell(a) = \log(1 + e^{-a})$, use the exact one-sample, all-layer update

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta}^{(\kappa)} \ell\left(h^*(x^{(t)}) f_{\theta^{(t)}}(x^{(t)})\right), \quad 0 \leq t < T. \quad (4)$$

The prescribed latter-half score, predictor, and strict classification error are

$$A_{\mathcal{D}, h^*}(x) := \sum_{t=\lceil T/2 \rceil}^T f_{\theta^{(t)}}(x), \quad \widehat{h}_{\mathcal{D}, h^*}(x) := \text{sign}_{s_0}(A_{\mathcal{D}, h^*}(x)), \quad (5)$$

$$\mathcal{L}_{\mathcal{D}, h^*}(\widehat{h}_{\mathcal{D}, h^*}) := \Pr_{x \sim \mathcal{D}} \left[\widehat{h}_{\mathcal{D}, h^*}(x) h^*(x) < 0 \right]. \quad (6)$$

For $r \geq 0$, let

$$B_{\infty}(\theta^{(0)}, r) := \{\theta \in \mathbb{R}^S : \|\theta - \theta^{(0)}\|_{\infty} \leq r\}. \quad (7)$$

Define the static hidden-preactivation margin

$$M_r(\theta^{(0)}) := \begin{cases} +\infty, & L = 1, \\ \inf_{\substack{\theta \in B_{\infty}(\theta^{(0)}, r), \, x \in \mathcal{X}, \\ 1 \leq \ell < L, \, j \in [n_{\ell}]} } |u_{\ell, j}(\theta, x)|, & L \geq 2, \end{cases} \quad (8)$$

and the exact selected-gradient envelope

$$G_r(\theta^{(0)}) := \sup_{\substack{\theta \in B_{\infty}(\theta^{(0)}, r), \, x \in \mathcal{X}, \\ y \in \{-1, +1\}}} \left\| \nabla_{\theta}^{(\kappa)} \ell(y f_{\theta}(x)) \right\|_{\infty}. \quad (9)$$

The initialization event used below is

$$E_r := \{M_r(\theta^{(0)}) > 0, \, \eta T G_r(\theta^{(0)}) \leq r\}. \quad (10)$$

It is determined solely by initialization and worst-case quantities on the fixed parameter ball, before a sample path, distribution, or target is selected.

Finally, introduce the path index set and its dimension

$$\mathcal{I}_{\text{path}} := \prod_{\ell=0}^{L-1} [n_{\ell}], \quad d_{\text{path}} := |\mathcal{I}_{\text{path}}| = \prod_{\ell=0}^{L-1} n_{\ell}. \quad (11)$$

For $p = (i_0, \dots, i_{L-1}) \in \mathcal{I}_{\text{path}}$, define the initialization-gate feature map by

$$[\varphi_{\theta^{(0)}}(x)]_p := x_{i_0} \prod_{\ell=1}^{L-1} \mathbf{1}\{u_{\ell, i_{\ell}}(\theta^{(0)}, x) > 0\}, \quad (12)$$

where the empty product is one for $L = 1$. Let $\mathcal{P}_{\text{gate}}$ be the pushforward law of $\varphi_{\theta^{(0)}}$ under the Gaussian initialization (3). This law is fixed by the architecture and initialization distribution before $\mathcal{D}$ and h , and it is defined on both E_r and E_r^c .

1.2 Assumptions

Assumption 1 (Fixed source witnesses and regime). *The parameters satisfy $0 \leq \varepsilon < 1/4$, $T \in \mathbb{N}_{\geq 1}$, and $\eta > 0$. For the given $(n, \mathcal{H}, \varepsilon)$, one architecture, its parameter count S , the constant stepsize η , and the horizon T are selected once and held fixed before $\mathcal{D}$ and h^* . The same witnesses and the same fixed (s_0, κ) conventions are used for every $(\mathcal{D}, h^*) \in \Delta(\mathcal{X}) \times \mathcal{H}$.*

Assumption 2 (Universal expected-error SGD premise). *For every $\mathcal{D} \in \Delta(\mathcal{X})$ and every $h^* \in \mathcal{H}$,*

$$\mathbb{E}_{\theta^{(0)}, x^{(0)}, \dots, x^{(T-1)}} \left[\mathcal{L}_{\mathcal{D}, h^*}(\hat{h}_{\mathcal{D}, h^*}) \right] \leq \varepsilon, \quad (13)$$

where the expectation is jointly over the independent Gaussian initialization and fresh one-sample SGD draws specified above.

Assumption 3 (Fixed constant depth). *A depth cap $L_0 \in \mathbb{N}_{\geq 1}$ is fixed as a universal constant, independently of $n, \mathcal{H}, \varepsilon, \eta, T, S$, and the source architecture satisfies $1 \leq L \leq L_0$. Widths remain arbitrary and all layers remain trainable.*

Assumption 4 (Static robust initialization tube). *Deterministic values $r > 0$ and δ_0 with $0 \leq \delta_0 \leq \varepsilon$ are fixed after the source witnesses but before $\theta^{(0)}, \mathcal{D}, h^*$, and the SGD samples, and*

$$\Pr_{\theta^{(0)}}(E_r) \geq 1 - \delta_0. \quad (14)$$

This is only a primitive statement about the static quantities M_r and G_r ; it does not assume trajectory containment, gate stability, or a path representation.

1.3 Formalized goal

The objective is the following conditional, fixed-depth probabilistic dimension statement:

$$\text{dc}_{\varepsilon + \delta_0}(\mathcal{H}) \leq d_{\text{path}} \leq S^{L_0}. \quad (15)$$

The first inequality must be witnessed by the single target- and distribution-independent law $\mathcal{P}_{\text{gate}}$; equivalently, for every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$,

$$\mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D}, h}(w, \varphi) \right] \leq \varepsilon + \delta_0. \quad (16)$$

Because $\delta_0 \leq \varepsilon$, the same witness should give

$$\text{dc}_{2\varepsilon}(\mathcal{H}) \leq S^{L_0}. \quad (17)$$

The statement is conditional on Assumption 4 and the constant-depth restriction. It does not assert the corresponding result in the absence of the robust tube, at general depth, or with a deterministic exact feature map.

2 Preliminaries

The coordinate norm $\|\cdot\|_\infty$ on $\mathbb{R}^S$ is used throughout because it is simultaneously the norm defining the static ball (7) and the norm in the gradient envelope (9). For a distribution $\mathcal{D}$ on the finite set $\mathcal{X}$, $\mathcal{D}^T$ denotes the law of the independent training tuple $(x^{(0)}, \dots, x^{(T-1)})$; it is used to keep the conditional sample averaging explicit. The law $\mathcal{P}_{\text{gate}}$ and dimension d_{path} retain the definitions (11)–(12), which are the public feature objects in the theorem below. All auxiliary activation envelopes, adjoints, history-dependent coefficients, and event-split functions are introduced only inside the appendix arguments where they are derived.

3 Main Theorem

Theorem 3.1 (Conditional polynomial probabilistic dimension). *Under Assumptions 1, 2, 3, and 4, the unconditional initialization-gate law $\mathcal{P}_{\text{gate}}$, selected before every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$, satisfies*

$$\mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \varphi) \right] \leq \varepsilon + \delta_0. \quad (18)$$

Consequently,

$$\text{dc}_{\varepsilon+\delta_0}(\mathcal{H}) \leq d_{\text{path}} \leq S^L \leq S^{L_0}, \quad (19)$$

and the same law is admissible at the larger threshold 2ε , so

$$\text{dc}_{2\varepsilon}(\mathcal{H}) \leq S^{L_0}. \quad (20)$$

These bounds are nonasymptotic and have no hidden multiplicative constant. The exposed quantitative variables are $S, L, L_0, \varepsilon, \delta_0, T, \eta, r$ through the assumptions and the exact conclusions above; L_0 is universally fixed. The learner premise is an expectation over initialization and the fixed finite sample horizon, the tube premise is an initialization probability, and (18) is an expectation over the unconditional law $\mathcal{P}_{\text{gate}}$. The metric is the exact tie-resolved classification risk, and no coefficient norm or positive score margin is required.

For $L = 1$, the feature map is the gate-free map $x \mapsto x$ and $d_{\text{path}} = S$. If $\delta_0 = 0$, the complement penalty disappears and (18) has right-hand side ε . If $\varepsilon = \delta_0 = 0$, both public error thresholds remain exactly zero. These reductions are part of the theorem rather than asymptotic limiting claims.

4 Proof Sketch

The proof first controls the exact protocol-selected gradient on the whole static coordinate ball. Lemma A.1 bounds all forward coordinates, and Proposition A.1 uses the fixed selector only through $0 \leq \rho_\kappa \leq 1$ to show that $G_r(\theta^{(0)})$ is finite. Given an initialization in E_r , Lemma A.2 and Proposition A.2 then induct over every possible labeled history and prove $\|\theta^{(t)} - \theta^{(0)}\|_\infty \leq t\eta G_r \leq r$ through time T .

The positive static margin M_r is converted into an exact structural statement, rather than a perturbative error. Lemma A.3 gives continuous preactivation paths inside the convex ball, and Proposition A.3 rules out sign crossings. Proposition A.4 transfers the resulting initialization gate signature to every generated state. Lemma A.4 expands each network output exactly in the single feature map $\varphi_{\theta^{(0)}}$, while Proposition A.5 sums the moving-weight coefficients over the prescribed latter half of the trajectory and preserves zero-score ties.

For fixed $\mathcal{D}, h$, and a stable initialization, Lemma A.5 identifies the path-linear and learner risks for every training tuple. Proposition A.6 compares the fixed-feature infimum with each history-dependent feasible coefficient before averaging, so there is no interchange of an infimum and an expectation. Lemma A.6 proves that the gate map has finite support and gives the exact pushforward expectation identity. Proposition A.7 applies the stable comparison on E_r , charges E_r^c by its unit risk and mass δ_0 , and invokes universal expected success. Proposition A.8 records that the law was fixed before the arbitrary distribution and target.

Finally, Lemma A.7 proves $d_{\text{path}} \leq S^L \leq S^{L_0}$ factor by factor, and Proposition A.9 reuses the same law after the exact threshold comparison $\varepsilon + \delta_0 \leq 2\varepsilon$. The final appendix proof combines these named results and no additional lemma.

A Proof Details

A.1 Finite gradient control on the static ball

Lemma A.1 (Finite-ball forward envelope). *Under Assumptions 1 and 4, fix a deterministic $\theta^{(0)} \in \mathbb{R}^S$ and the finite radius $r > 0$. Define*

$$B := \max\{1, \|\theta^{(0)}\|_\infty + r\}, \quad a_0 := 1, \quad a_\ell := n_{\ell-1} B a_{\ell-1} \quad (1 \leq \ell < L). \quad (21)$$

Then $B, a_0, \dots, a_{L-1}$ are finite and, for every $\theta \in B_\infty(\theta^{(0)}, r)$, $x \in \mathcal{X}$, hidden layer $1 \leq \ell < L$, and $j \in [n_\ell]$,

$$|u_{\ell,j}(\theta, x)| \leq a_\ell, \quad |z_{\ell,j}(\theta, x)| \leq a_\ell. \quad (22)$$

For $L = 1$ the hidden-layer assertion is empty and $a_0 = 1$.

Proof. All coordinates of the deterministic vector $\theta^{(0)}$ are finite and the radius is finite, hence $B < \infty$. If θ lies in the closed ball, then every weight coordinate satisfies

$$|\theta_{\ell,jk}| \leq |(\theta_\ell^{(0)})_{jk}| + |\theta_{\ell,jk} - (\theta_\ell^{(0)})_{jk}| \leq \|\theta^{(0)}\|_\infty + r \leq B. \quad (23)$$

The weak inequalities include the boundary $\|\theta - \theta^{(0)}\|_\infty = r$. At the input, $z_0(\theta, x) = x$ and every coordinate has magnitude $1 = a_0$. If all coordinates of $z_{\ell-1}$ have magnitude at most $a_{\ell-1}$, then

$$\begin{aligned} |u_{\ell,j}(\theta, x)| &= \left| \sum_{k=1}^{n_{\ell-1}} \theta_{\ell,jk} z_{\ell-1,k}(\theta, x) \right| \\ &\leq \sum_{k=1}^{n_{\ell-1}} |\theta_{\ell,jk}| |z_{\ell-1,k}(\theta, x)| \\ &\leq n_{\ell-1} B a_{\ell-1} = a_\ell. \end{aligned} \quad (24)$$

Since $0 \leq \sigma(v) \leq |v|$ for every real v , the same bound holds for $z_{\ell,j}$. Finite induction proves (22), and the recurrence also gives

$$a_\ell = B^\ell \prod_{q=0}^{\ell-1} n_q, \quad (25)$$

with the empty product equal to one. Thus all envelopes are finite. $\square$

Proposition A.1 (Exact fixed-selector gradient envelope). *Under Assumptions 1 and 4, and Lemma A.1, define*

$$b_L := 1, \quad b_\ell := n_{\ell+1} B b_{\ell+1} \quad (1 \leq \ell < L), \quad C_r(\theta^{(0)}) := \max_{1 \leq \ell \leq L} b_\ell a_{\ell-1}. \quad (26)$$

Then on the full domain defining $G_r(\theta^{(0)})$,

$$\left| [\nabla_\theta^{(\kappa)} \ell(y f_\theta(x))]_{\ell,jk} \right| \leq b_\ell a_{\ell-1} \leq C_r(\theta^{(0)}), \quad (27)$$

for every layer and coordinate. Consequently

$$G_r(\theta^{(0)}) \leq C_r(\theta^{(0)}) < \infty. \quad (28)$$

When $L = 1$, $C_r(\theta^{(0)}) = 1$ and $G_r(\theta^{(0)}) \leq 1$.

Proof. Fix θ, x, y in the domains of (27). The forward equations are finite sums and products, so $f_\theta(x)$ is finite. Direct differentiation of $\ell(a) = \log(1 + e^{-a})$ gives

$$\ell'(a) = -\frac{1}{1 + e^a}, \quad \left| \frac{\partial}{\partial f} \ell(yf) \right| = \frac{1}{1 + e^{yf}} \leq 1. \quad (29)$$

Because $n_L = 1$, define the selected output adjoint

$$\delta_{L,1} := \frac{\partial}{\partial f} \ell(yf) \Big|_{f=f_\theta(x)} = -\frac{y}{1 + e^{yf_\theta(x)}}. \quad (30)$$

Thus $|\delta_{L,1}| \leq 1 = b_L$. The exact protocol-selected recurrence at the hidden layers is

$$\delta_{\ell,j} := \rho_\kappa(u_{\ell,j}(\theta, x)) \sum_{q=1}^{n_{\ell+1}} \theta_{\ell+1,qj} \delta_{\ell+1,q}, \quad \ell = L-1, \dots, 1. \quad (31)$$

It is valid as the definition of the selected back-propagation value even at a kink. Since $0 \leq \rho_\kappa(v) \leq 1$ for every v , including $\rho_\kappa(0) = \kappa$, the induction hypothesis and (23) give

$$|\delta_{\ell,j}| \leq \sum_{q=1}^{n_{\ell+1}} |\theta_{\ell+1,qj}| |\delta_{\ell+1,q}| \leq n_{\ell+1} B b_{\ell+1} = b_\ell. \quad (32)$$

Backward induction yields

$$b_\ell = B^{L-\ell} \prod_{q=\ell+1}^L n_q, \quad (33)$$

where $n_L = 1$. The layerwise selected-gradient formula is

$$[\nabla_\theta^{(\kappa)} \ell(yf_\theta(x))]_{\ell,jk} = \delta_{\ell,j} z_{\ell-1,k}(\theta, x). \quad (34)$$

Combining (32), (34), and Lemma A.1 proves (27), and taking the supremum over the domain in (9) proves (28). The finiteness of (25), (33), and the finite layer maximum in (26) gives $C_r < \infty$. The estimate includes hidden preactivation zeros for every $\kappa \in [0, 1]$ and does not use continuity or attainment of the selected gradient supremum.

If $L = 1$, there are no hidden adjoints and (34) reduces to

$$\left| [\nabla_\theta^{(\kappa)} \ell(yf_\theta(x))]_{1,1k} \right| = \frac{|x_k|}{1 + e^{yf_\theta(x)}} \leq 1, \quad (35)$$

so $a_0 = b_1 = C_r = 1$, including zero weights and closed-ball boundary points. $\square$

The two results prove the finite static envelope before any trajectory membership or gate statement is invoked. The bound is deterministic for each initialization and uniform in inputs and labels.

A.2 All-history containment in the closed tube

Fix $\theta^{(0)} \in E_r$ and an arbitrary deterministic labeled history

$$\omega = ((x^{(0)}, y^{(0)}), \dots, (x^{(T-1)}, y^{(T-1)})) \in (\mathcal{X} \times \{-1, +1\})^T.$$

Let the exact history-driven states satisfy

$$\theta^{(t+1)} = \theta^{(t)} - \eta g_t, \quad g_t := \nabla_\theta^{(\kappa)} \ell(y^{(t)} f_{\theta^{(t)}}(x^{(t)})), \quad 0 \leq t < T, \quad (36)$$

and put

$$D_t := \|\theta^{(t)} - \theta^{(0)}\|_\infty, \quad G := G_r(\theta^{(0)}). \quad (37)$$

Lemma A.2 (One-step closed-ball recurrence). *Under Assumptions 1 and 4, Proposition A.1, and the definitions (36)–(37), if $0 \leq t < T$ and $D_t \leq r$, then*

$$\|g_t\|_\infty \leq G, \quad D_{t+1} \leq D_t + \eta G. \quad (38)$$

The conclusion includes the case $D_t = r$.

Proof. The condition $D_t \leq r$ is exactly $\theta^{(t)} \in B_\infty(\theta^{(0)}, r)$. Since the history is arbitrary, $x^{(t)} \in \mathcal{X}$ and $y^{(t)} \in \{-1, +1\}$. Therefore the triple $(\theta^{(t)}, x^{(t)}, y^{(t)})$ lies in the defining domain of G_r , and

$$\|g_t\|_\infty = \left\| \nabla_\theta^{(\kappa)} \ell(y^{(t)} f_{\theta^{(t)}}(x^{(t)})) \right\|_\infty \leq G. \quad (39)$$

Using (36), the triangle inequality, $\eta > 0$, and (39),

$$\begin{aligned} D_{t+1} &= \|\theta^{(t)} - \theta^{(0)} - \eta g_t\|_\infty \\ &\leq \|\theta^{(t)} - \theta^{(0)}\|_\infty + \eta \|g_t\|_\infty \\ &\leq D_t + \eta G. \end{aligned} \quad (40)$$

No sign, descent, or cancellation property is used. $\square$

Proposition A.2 (All-history closed-tube containment). *Under Assumptions 1 and 4, Proposition A.1, and Lemma A.2, fix any initialization $\theta^{(0)} \in E_r$ and any deterministic labeled history*

$$\omega = ((x^{(0)}, y^{(0)}), \dots, (x^{(T-1)}, y^{(T-1)})) \in (\mathcal{X} \times \{-1, +1\})^T.$$

Let $\theta^{(t)}$ be generated from $\theta^{(0)}$ by the exact recursion

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta^{(\kappa)} \ell(y^{(t)} f_{\theta^{(t)}}(x^{(t)})), \quad 0 \leq t < T.$$

Then, for every integer $0 \leq t \leq T$,

$$\|\theta^{(t)} - \theta^{(0)}\|_\infty \leq t\eta G_r(\theta^{(0)}) \leq r. \quad (41)$$

Proof. The accepted static envelope gives $0 \leq G < \infty$. Membership in E_r gives only the scalar budget

$$\eta TG \leq r. \quad (42)$$

At the entry state,

$$D_0 = \|\theta^{(0)} - \theta^{(0)}\|_\infty = 0 = 0 \cdot \eta G \leq r. \quad (43)$$

Suppose $0 \leq t < T$ and $D_t \leq t\eta G$. Since $t \leq T$, (42) gives

$$D_t \leq t\eta G \leq T\eta G \leq r, \quad (44)$$

so the one-step lemma may be applied only after current membership has been established. It yields

$$D_{t+1} \leq D_t + \eta G \leq (t+1)\eta G, \quad (45)$$

and $t+1 \leq T$ gives

$$(t+1)\eta G \leq T\eta G \leq r. \quad (46)$$

Finite induction proves (41). The first transition is $D_1 \leq \eta G \leq r$ because $T \geq 1$; if $T = 1$, (43) and this bound complete the argument. If $G = 0$, every inequality forces $D_t = 0$. Equality at the terminal radius is allowed because the ball is closed, and arbitrary outward increments are already charged by the full sum $t\eta G$. $\square$

Thus the static event supplies no trajectory containment statement by itself: (41) is the derived all-history containment result that will be used below. It applies in particular to source-consistent histories with $y^{(t)} = h(x^{(t)})$.

A.3 Gate constancy from the positive margin

Lemma A.3 (Continuous exact preactivation paths). *Under Assumptions 1 and 4, fix $\theta^{(0)}$, $r > 0$, $\theta \in B_\infty(\theta^{(0)}, r)$, and $x \in \mathcal{X}$. For $L \geq 2$ define*

$$\gamma(s) := \theta^{(0)} + s(\theta - \theta^{(0)}), \quad 0 \leq s \leq 1. \quad (47)$$

Then $\gamma(s)$ remains in the closed ball and each exact scalar path $q_{\ell,j}(s) := u_{\ell,j}(\gamma(s), x)$, $1 \leq \ell < L$, is continuous. More explicitly,

$$q_{1,j}(s) = \sum_{k=1}^{n_0} \gamma_{1,jk}(s) x_k, \quad (48)$$

and, for $2 \leq \ell < L$,

$$q_{\ell,j}(s) = \sum_{k=1}^{n_{\ell-1}} \gamma_{\ell,jk}(s) \sigma(q_{\ell-1,k}(s)). \quad (49)$$

For $L = 1$ there are no such paths.

Proof. For every vectorized coordinate,

$$|\gamma_a(s) - \theta_a^{(0)}| = s|\theta_a - \theta_a^{(0)}| \leq sr \leq r, \quad (50)$$

so the segment, including its endpoints and boundary points, lies in the closed ball. Its coordinates are affine and hence continuous. Formula (48) is a finite sum of continuous functions. If all paths at layer $\ell - 1$ are continuous, then coordinatewise ReLU is continuous, and (49) is a finite sum of products of continuous functions. Induction over the finite hidden layers proves continuity. These are the exact forward equations evaluated on $\gamma(s)$, not a linearization, and no derivative convention is used. $\square$

Proposition A.3 (Strict gate constancy on the positive-margin ball). *Under Assumptions 1 and 4, Lemma A.3, and the local condition $\theta^{(0)} \in E_r$, let $m = M_r(\theta^{(0)})$. If $L \geq 2$, then $m > 0$ and, for every point in the closed ball, input, hidden layer, and coordinate,*

$$|u_{\ell,j}(\theta, x)| \geq m, \quad \text{sgn}(u_{\ell,j}(\theta, x)) = \text{sgn}(u_{\ell,j}(\theta^{(0)}, x)), \quad (51)$$

where ordinary sgn is used only on nonzero reals. Hence

$$\mathbf{1}\{u_{\ell,j}(\theta, x) > 0\} = \mathbf{1}\{u_{\ell,j}(\theta^{(0)}, x) > 0\}. \quad (52)$$

For $L = 1$ the assertion is empty and $M_r = +\infty$ by definition.

Proof. When $L \geq 2$, event membership gives $m > 0$. By the infimum definition of M_r , every point ϑ in the ball satisfies

$$|u_{\ell,j}(\vartheta, x)| \geq m \quad (53)$$

for every hidden coordinate and input. Along the segment (47), (53) gives $|q_{\ell,j}(s)| \geq m > 0$. If the endpoint signs differed, continuity from Lemma A.3 and the intermediate value theorem would give an $s_\star \in (0, 1)$ with $q_{\ell,j}(s_\star) = 0$, contradicting (53). Thus the endpoint signs agree; since both are nonzero, (52) follows.

The argument includes $\theta = \theta^{(0)}$, all ball-boundary points, and zero parameter coordinates. A zero preactivation anywhere in the ball would make $M_r = 0$ and hence place the initialization outside E_r ; that case is not silently excluded from the later unconditional law. The proof never evaluates $\rho_\kappa(0)$ and is independent of κ . $\square$

Proposition A.4 (All-history initialization gate signature). *Under Assumptions 1 and 4, Proposition A.2, and Proposition A.3, fix any initialization $\theta^{(0)} \in E_r$ and any deterministic labeled history*

$$\omega = ((x^{(0)}, y^{(0)}), \dots, (x^{(T-1)}, y^{(T-1)})) \in (\mathcal{X} \times \{-1, +1\})^T.$$

Let $\theta^{(t)}$ be generated from $\theta^{(0)}$ by

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta}^{(\kappa)} \ell(y^{(t)} f_{\theta^{(t)}}(x^{(t)})), \quad 0 \leq t < T.$$

If $L \geq 2$, then

$$|u_{\ell,j}(\theta^{(t)}, x)| \geq M_r(\theta^{(0)}) > 0, \quad \mathbf{1}\{u_{\ell,j}(\theta^{(t)}, x) > 0\} = \mathbf{1}\{u_{\ell,j}(\theta^{(0)}, x) > 0\} \quad (54)$$

for all x , hidden (ℓ, j) , and $0 \leq t \leq T$. For $L = 1$ the universal statement is empty.

Proof. Proposition A.2 gives

$$\|\theta^{(t)} - \theta^{(0)}\|_{\infty} \leq t\eta G_r(\theta^{(0)}) \leq r \quad (55)$$

for every state, including the first and terminal states and equality at the boundary. Proposition A.3 therefore applies to each $\theta^{(t)}$, every input, and every hidden coordinate, proving (54). The history was arbitrary, so the conclusion is all-history rather than almost-sure. No gate stability was assumed before (55) was established. $\square$

A.4 Exact path expansion and aggregate coefficients

Lemma A.4 (Exact path expansion under the initialization gate signature). *Under Assumptions 1 and 4, Proposition A.3, and $\theta^{(0)} \in E_r$, fix $\theta \in B_{\infty}(\theta^{(0)}, r)$ and $x \in \mathcal{X}$. For $p = (i_0, \dots, i_{L-1}) \in \mathcal{I}_{\text{path}}$ define*

$$c_p(\theta) := (\theta_L)_{1, i_{L-1}} \prod_{\ell=1}^{L-1} (\theta_{\ell})_{i_{\ell}, i_{\ell-1}}, \quad \varphi_{\theta^{(0)}, p}(x) := x_{i_0} \prod_{\ell=1}^{L-1} \mathbf{1}\{u_{\ell, i_{\ell}}(\theta^{(0)}, x) > 0\}, \quad (56)$$

and let $c(\theta) = (c_p(\theta))_{p \in \mathcal{I}_{\text{path}}}$. Then

$$f_{\theta}(x) = \sum_{p \in \mathcal{I}_{\text{path}}} c_p(\theta) \varphi_{\theta^{(0)}, p}(x) = \langle c(\theta), \varphi_{\theta^{(0)}}(x) \rangle. \quad (57)$$

Empty products are one, so this includes $L = 1$.

Proof. Assume first $L \geq 2$ and abbreviate

$$g_{\ell,j}^{(0)}(x) := \mathbf{1}\{u_{\ell,j}(\theta^{(0)}, x) > 0\}. \quad (58)$$

The ball gate proposition gives

$$\mathbf{1}\{u_{\ell,j}(\theta, x) > 0\} = g_{\ell,j}^{(0)}(x), \quad |u_{\ell,j}(\theta, x)| > 0. \quad (59)$$

Therefore the exact ReLU equation is

$$z_{\ell,j}(\theta, x) = g_{\ell,j}^{(0)}(x) \sum_{k=1}^{n_{\ell-1}} (\theta_{\ell})_{j,k} z_{\ell-1,k}(\theta, x). \quad (60)$$

Define finite partial path sums by

$$P_{0,i_0}(\theta, x) := x_{i_0}, \quad (61)$$

and, for $1 \leq \ell < L$,

$$P_{\ell, i_\ell}(\theta, x) := g_{\ell, i_\ell}^{(0)}(x) \sum_{i_{\ell-1}=1}^{n_{\ell-1}} (\theta_\ell)_{i_\ell, i_{\ell-1}} P_{\ell-1, i_{\ell-1}}(\theta, x). \quad (62)$$

We prove simultaneously $z_{\ell, j} = P_{\ell, j}$ and, for every hidden layer, the explicit expansion

$$P_{\ell, i_\ell}(\theta, x) = \sum_{i_0=1}^{n_0} \cdots \sum_{i_{\ell-1}=1}^{n_{\ell-1}} x_{i_0} \left(\prod_{q=1}^{\ell} g_{q, i_q}^{(0)}(x) \right) \left(\prod_{q=1}^{\ell} (\theta_q)_{i_q, i_{q-1}} \right). \quad (63)$$

At layer zero, $z_{0, i_0} = x_{i_0} = P_{0, i_0}$. If the assertion holds at layer $\ell - 1$, substituting it into (60) gives (62), and distributing the finite sum appends exactly the new factor $g_{\ell, i_\ell}^{(0)}(x)(\theta_\ell)_{i_\ell, i_{\ell-1}}$, proving (63). No division or nonzero weight condition is used.

At the scalar output, $n_L = 1$ and

$$\begin{aligned} f_\theta(x) &= \sum_{i_0=1}^{n_0} \cdots \sum_{i_{L-1}=1}^{n_{L-1}} x_{i_0} \left(\prod_{q=1}^{L-1} g_{q, i_q}^{(0)}(x) \right) \left[(\theta_L)_{1, i_{L-1}} \prod_{q=1}^{L-1} (\theta_q)_{i_q, i_{q-1}} \right] \\ &= \sum_{p \in \mathcal{I}_{\text{path}}} c_p(\theta) \varphi_{\theta^{(0)}, p}(x). \end{aligned} \quad (64)$$

If $L = 1$, then $\mathcal{I}_{\text{path}} = [n_0]$ and the original output equation is already

$$f_\theta(x) = \sum_{i_0=1}^{n_0} (\theta_1)_{1, i_0} x_{i_0} = \sum_{i_0 \in [n_0]} c_{i_0}(\theta) \varphi_{\theta^{(0)}, i_0}(x). \quad (65)$$

Zero or canceling products simply give zero summands or exact cancellation; the finite equality remains valid. $\square$

Proposition A.5 (Exact latter-half aggregate coefficient and tie identity). *Under Assumptions 1 and 4, Propositions A.2 and A.4, and Lemma A.4, fix an initialization in E_r and a deterministic labeled history with its exact states. Set*

$$J_T := \{\lceil T/2 \rceil, \dots, T\}, \quad w_\omega := \sum_{t \in J_T} c(\theta^{(t)}(\omega)). \quad (66)$$

Then, for every $x \in \mathcal{X}$,

$$A_\omega(x) := \sum_{t \in J_T} f_{\theta^{(t)}(\omega)}(x) = \langle w_\omega, \varphi_{\theta^{(0)}}(x) \rangle. \quad (67)$$

For a source-consistent history this is the setting score, and applying sign_{s_0} gives the same prediction, including at a zero score.

Proof. Proposition A.2 places each state in the closed ball, and Proposition A.4 supplies the initialization gate signature at every state. Lemma A.4 therefore gives, for each $t \in J_T$,

$$f_{\theta^{(t)}(\omega)}(x) = \left\langle c(\theta^{(t)}(\omega)), \varphi_{\theta^{(0)}}(x) \right\rangle. \quad (68)$$

The set J_T is finite and nonempty because $T \geq 1$. Summing (68) and using bilinearity proves (67). On a source-consistent history the left side is exactly (5), so equal real scores have equal sign_{s_0} values and identical strict-error indicators. When $T = 1$, $J_T = \{1\}$. Zero coefficients, cancellations, and a common score of zero do not change the equality; at zero both predictors use s_0 . $\square$

A.5 The conditional infimum comparison

Fix $\mathcal{D} \in \Delta(\mathcal{X})$, $h \in \mathcal{H}$, and a stable initialization $\theta^{(0)} \in E_r$. For a training tuple $\mathbf{x} = (x^{(0)}, \dots, x^{(T-1)}) \in \mathcal{X}^T$, use the source-consistent history

$$\omega_h(\mathbf{x}) := ((x^{(0)}, h(x^{(0)})), \dots, (x^{(T-1)}, h(x^{(T-1)}))). \quad (69)$$

Let $w_{\mathbf{x}}$ be the coefficient in Proposition A.5, and abbreviate the corresponding score and predictor by

$$A_{\theta^{(0)}, \mathbf{x}}(x) := \sum_{t=\lceil T/2 \rceil}^T f_{\theta^{(t)}(\theta^{(0)}, \omega_h(\mathbf{x}))}(x), \quad \hat{h}_{\theta^{(0)}, \mathbf{x}}(x) := \text{sign}_{s_0}(A_{\theta^{(0)}, \mathbf{x}}(x)). \quad (70)$$

Lemma A.5 (Pathwise exact risk identity). *Under Assumptions 1 and 4, Proposition A.5, and fixed $\mathcal{D}, h, \theta^{(0)} \in E_r$, every $\mathbf{x} \in \mathcal{X}^T$ has a vector $w_{\mathbf{x}} \in \mathbb{R}^{d_{\text{path}}}$ such that*

$$\text{sign}_{s_0}(A_{\theta^{(0)}, \mathbf{x}}(x)) = \text{sign}_{s_0}(\langle w_{\mathbf{x}}, \varphi_{\theta^{(0)}}(x) \rangle) \quad (71)$$

for every x , and

$$R_{\mathcal{D}, h}(w_{\mathbf{x}}, \varphi_{\theta^{(0)}}) = \mathcal{L}_{\mathcal{D}, h}(\hat{h}_{\theta^{(0)}, \mathbf{x}}). \quad (72)$$

Proof. The history (69) is deterministic and source-consistent. Proposition A.5 gives the real-score equality

$$A_{\theta^{(0)}, \mathbf{x}}(x) = \langle w_{\mathbf{x}}, \varphi_{\theta^{(0)}}(x) \rangle \quad (73)$$

for every evaluation input. Applying the same tie-resolved sign, multiplying by the same $h(x)$, and testing the same strict inequality gives pointwise equality of the error indicators, hence (72). The conclusion remains exact when the common score is zero, and no coefficient nonzero or margin condition is needed. For $T = 1$, the index set contains the single terminal state. $\square$

Proposition A.6 (Conditional infimum-before-expectation comparison). *Under Assumptions 1 and 4, Proposition A.5, and Lemma A.5, fix $\mathcal{D} \in \Delta(\mathcal{X})$, $h \in \mathcal{H}$, and $\theta^{(0)} \in E_r$ before drawing $\mathbf{X} \sim \mathcal{D}^T$. Define*

$$F_{\mathcal{D}, h}(\theta^{(0)}) := \inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D}, h}(w, \varphi_{\theta^{(0)}}). \quad (74)$$

Then

$$0 \leq F_{\mathcal{D}, h}(\theta^{(0)}) \leq 1, \quad F_{\mathcal{D}, h}(\theta^{(0)}) \leq \mathbb{E}_{\mathbf{X} \sim \mathcal{D}^T} [\mathcal{L}_{\mathcal{D}, h}(\hat{h}_{\theta^{(0)}, \mathbf{X}})]. \quad (75)$$

The infimum need not be attained.

Proof. Every risk in (74) is a probability in $[0, 1]$, and the coefficient space is nonempty, so its infimum is a finite value in $[0, 1]$. For every fixed $\mathbf{x}$, the pathwise lemma supplies the feasible vector $w_{\mathbf{x}}$; the defining lower-bound property of an infimum gives

$$F_{\mathcal{D}, h}(\theta^{(0)}) \leq R_{\mathcal{D}, h}(w_{\mathbf{x}}, \varphi_{\theta^{(0)}}) = \mathcal{L}_{\mathcal{D}, h}(\hat{h}_{\theta^{(0)}, \mathbf{x}}). \quad (76)$$

The left side is fixed before the tuple is drawn. Multiplying (76) by the nonnegative mass $\mathcal{D}^T(\mathbf{x})$ and summing over the finite set $\mathcal{X}^T$ gives

$$\begin{aligned} \mathbb{E}_{\mathbf{X} \sim \mathcal{D}^T} [\mathcal{L}_{\mathcal{D}, h}(\hat{h}_{\theta^{(0)}, \mathbf{X}})] &= \sum_{\mathbf{x} \in \mathcal{X}^T} \mathcal{D}^T(\mathbf{x}) \mathcal{L}_{\mathcal{D}, h}(\hat{h}_{\theta^{(0)}, \mathbf{x}}) \\ &\geq \sum_{\mathbf{x} \in \mathcal{X}^T} \mathcal{D}^T(\mathbf{x}) F_{\mathcal{D}, h}(\theta^{(0)}) \\ &= F_{\mathcal{D}, h}(\theta^{(0)}). \end{aligned} \quad (77)$$

This is pointwise feasible-candidate comparison followed by averaging, not an interchange of an infimum with an expectation, and it does not require a measurable minimizing coefficient. $\square$

A.6 Unconditional gate-law witness

Let $\Theta = \mathbb{R}^S$ and let γ be the product Gaussian initialization law in (3). Define

$$\Phi : \Theta \longrightarrow \{\text{maps } \mathcal{X} \rightarrow \mathbb{R}^{d_{\text{path}}}\}, \quad \Phi(\vartheta) := \varphi_{\vartheta}, \quad (78)$$

and, for fixed $\mathcal{D}, h$,

$$F_{\mathcal{D},h}(\vartheta) := \inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \Phi(\vartheta)). \quad (79)$$

This definition applies on both the stable event and its complement.

Lemma A.6 (Finite gate law and pushforward identity). *Under Assumption 1, the map Φ is Borel measurable and has finite range $\Gamma = \Phi(\Theta)$, with*

$$|\Gamma| \leq 3^{d_{\text{path}}|\mathcal{X}|}. \quad (80)$$

For every fixed $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$, $F_{\mathcal{D},h}$ is Borel, takes values in $[0, 1]$, and

$$\mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \varphi) \right] = \mathbb{E}_{\theta^{(0)} \sim \gamma} [F_{\mathcal{D},h}(\theta^{(0)})]. \quad (81)$$

No optimizing coefficient is selected measurably.

Proof. For fixed x , every initialization preactivation is continuous in ϑ by finite matrix multiplication and continuous coordinatewise ReLU. Hence each strict indicator $\mathbf{1}\{u_{\ell,j}(\vartheta, x) > 0\}$ is Borel. Each coordinate of Φ is a finite product of such indicators and one input coordinate, and therefore belongs to $\{-1, 0, +1\}$. There are only $d_{\text{path}}|\mathcal{X}|$ coordinates, proving measurability and (80). For $L = 1$, the empty product gives $\Phi(\vartheta)(x) = x$ and the range is a singleton.

For $\psi \in \Gamma$, define the finite-table function

$$\Psi_{\mathcal{D},h}(\psi) := \inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \psi). \quad (82)$$

Every risk is in $[0, 1]$ and the coefficient space is nonempty, so the table is finite-valued even if its infimum is not attained. Equation (79) is $F_{\mathcal{D},h} = \Psi_{\mathcal{D},h} \circ \Phi$, hence F is Borel. Since $\mathcal{P}_{\text{gate}} = \Phi_{\#}\gamma$, summing over the finite range gives

$$\begin{aligned} \mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_w R_{\mathcal{D},h}(w, \varphi) \right] &= \sum_{\psi \in \Gamma} \Psi_{\mathcal{D},h}(\psi) \gamma(\Phi^{-1}(\{\psi\})) \\ &= \int_{\Theta} \Psi_{\mathcal{D},h}(\Phi(\vartheta)) d\gamma(\vartheta) \\ &= \mathbb{E}_{\theta^{(0)} \sim \gamma} [F_{\mathcal{D},h}(\theta^{(0)})]. \end{aligned} \quad (83)$$

Initializations inducing the same map are combined by the pushforward mass; no coefficient selector appears. $\square$

Proposition A.7 (Unconditional event-split risk bound). *Under Assumptions 1, 2, and 4, and Proposition A.6, for every fixed $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$,*

$$\mathbb{E}_{\theta^{(0)} \sim \gamma} [F_{\mathcal{D},h}(\theta^{(0)})] \leq \varepsilon + \delta_0. \quad (84)$$

The expectation is under the unconditional Gaussian law.

Proof. For a fixed pair $(x, y) \in \mathcal{X} \times \{-1, +1\}$, define the full one-step update map

$$\mathsf{T}_{x,y}(\theta) := \theta - \eta \nabla_{\theta}^{(\kappa)} \ell(y f_{\theta}(x)), \quad \theta \in \Theta.$$

This map is Borel. Indeed, every forward preactivation and activation is continuous in θ . The selected gate factor $\rho_{\kappa}(u_{\ell,j}(\theta, x))$ is Borel, including at zero; the output adjoint $-y/(1 + e^{y f_{\theta}(x)})$ is continuous; and the finite backward recursion and the coordinate formula $[\nabla_{\theta}^{(\kappa)} \ell(y f_{\theta}(x))]_{\ell,jk} = \delta_{\ell,j} z_{\ell-1,k}(\theta, x)$ use only finite sums and products. Thus the selected gradient is a Borel map from Θ to Θ , and subtracting it from the identity proves the claim for $\mathsf{T}_{x,y}$.

Fix a tuple $\mathbf{x} = (x^{(0)}, \dots, x^{(T-1)}) \in \mathcal{X}^T$ and define the generated-state maps recursively by

$$\mathsf{U}_{\mathbf{x},0}(\vartheta) := \vartheta, \quad \mathsf{U}_{\mathbf{x},t+1}(\vartheta) := \mathsf{T}_{x^{(t)}, h(x^{(t)})}(\mathsf{U}_{\mathbf{x},t}(\vartheta)), \quad 0 \leq t < T.$$

The base map is the identity. If $\mathsf{U}_{\mathbf{x},t}$ is Borel, then the displayed composition with the Borel map $\mathsf{T}_{x^{(t)}, h(x^{(t)})}$ is Borel. Finite induction therefore proves that every generated state map $\mathsf{U}_{\mathbf{x},t} : \Theta \rightarrow \Theta$ is Borel for $0 \leq t \leq T$.

For each evaluation input $v \in \mathcal{X}$, the finite-horizon score and predictor are consequently the Borel compositions

$$A_{\vartheta, \mathbf{x}}(v) := \sum_{t=\lceil T/2 \rceil}^T f_{\mathsf{U}_{\mathbf{x},t}(\vartheta)}(v), \quad \hat{h}_{\vartheta, \mathbf{x}}(v) := \text{sign}_{s_0}(A_{\vartheta, \mathbf{x}}(v)).$$

The tie map is Borel. Hence, for an initialization ϑ and the fixed tuple $\mathbf{x}$, the prescribed learner loss is the finite sum

$$Z_{\mathcal{D},h}(\vartheta, \mathbf{x}) := \mathcal{L}_{\mathcal{D},h}(\hat{h}_{\vartheta, \mathbf{x}}) = \sum_{v \in \mathcal{X}} \mathcal{D}(v) \mathbf{1}\{\text{sign}_{s_0}(A_{\vartheta, \mathbf{x}}(v))h(v) < 0\} \in [0, 1] \quad (85)$$

on the source-consistent labeled tuple $((x^{(t)}, h(x^{(t)}))_{t=0}^{T-1})$. It is Borel in ϑ for each fixed tuple. Since $\mathcal{X}^T$ is finite with its discrete sigma-algebra, $Z_{\mathcal{D},h}$ is jointly measurable on $\Theta \times \mathcal{X}^T$. Define the conditional sample average

$$g_{\mathcal{D},h}(\vartheta) := \mathbb{E}_{\mathbf{X} \sim \mathcal{D}^T} [Z_{\mathcal{D},h}(\vartheta, \mathbf{X})] = \sum_{\mathbf{x} \in \mathcal{X}^T} \mathcal{D}^T(\mathbf{x}) Z_{\mathcal{D},h}(\vartheta, \mathbf{x}), \quad (86)$$

which is measurable and lies in $[0, 1]$.

For every $\vartheta \in E_r$, the conditional comparison gives

$$F_{\mathcal{D},h}(\vartheta) \leq g_{\mathcal{D},h}(\vartheta). \quad (87)$$

For every $\vartheta \in E_r^c$, the probability definition gives

$$0 \leq F_{\mathcal{D},h}(\vartheta) \leq 1. \quad (88)$$

Splitting the same unconditional expectation therefore yields

$$\begin{aligned} \mathbb{E}_{\theta^{(0)}} F_{\mathcal{D},h}(\theta^{(0)}) &= \mathbb{E}[\mathbf{1}_{E_r} F_{\mathcal{D},h}] + \mathbb{E}[\mathbf{1}_{E_r^c} F_{\mathcal{D},h}] \\ &\leq \mathbb{E}[\mathbf{1}_{E_r} g_{\mathcal{D},h}] + \Pr(E_r^c). \end{aligned} \quad (89)$$

By independence and the finite sum (86), bounded tower averaging gives

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{E_r} g_{\mathcal{D},h}] &= \mathbb{E}_{\theta^{(0)}, \mathbf{X}}[\mathbf{1}_{E_r} Z_{\mathcal{D},h}(\theta^{(0)}, \mathbf{X})] \\ &\leq \mathbb{E}_{\theta^{(0)}, \mathbf{X}}[Z_{\mathcal{D},h}(\theta^{(0)}, \mathbf{X})] \\ &\leq \varepsilon, \end{aligned} \quad (90)$$

where the last inequality is Assumption 2. Assumption 4 gives

$$\Pr(E_r^c) \leq \delta_0. \quad (91)$$

Substitution in (89) proves (84). The complement is charged once by its unit risk bound; it is neither discarded nor used to condition the feature law. $\square$

Proposition A.8 (Universal probabilistic-dimension witness). *Under Assumptions 1, 2, and 4, together with Lemma A.6 and Proposition A.7, the one law $\mathcal{P}_{\text{gate}}$ satisfies, for every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$,*

$$\mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \varphi) \right] \leq \varepsilon + \delta_0, \quad (92)$$

and hence

$$\text{dc}_{\varepsilon+\delta_0}(\mathcal{H}) \leq d_{\text{path}}. \quad (93)$$

Proof. The architecture, Gaussian law, and rule $\Phi(\vartheta) = \varphi_{\vartheta}$ are fixed by Assumption 1 before any distribution or target. Thus their pushforward $\mathcal{P}_{\text{gate}}$ is one law independent of later choices. For arbitrary $\mathcal{D}, h$, (81) and (84) give (92). Since the law was fixed before this arbitrary choice, the quantifier order is $\exists \mathcal{P}_{\text{gate}} \forall \mathcal{D} \forall h$. The definition (1) then gives (93). $\square$

A.7 Polynomial dimension and the two-epsilon specialization

Lemma A.7 (Polynomial path-count bound). *Under Assumption 3, positive integer widths $n_0, \dots, n_L$ with $n_L = 1$, and the setting definitions $S = \sum_{j=1}^L n_j n_{j-1}$ and $d_{\text{path}} = \prod_{j=0}^{L-1} n_j$,*

$$d_{\text{path}} \leq S^L \leq S^{L_0}. \quad (94)$$

Proof. For each $j \in \{0, \dots, L-1\}$, positivity of n_{j+1} gives

$$n_j \leq n_{j+1} n_j \leq \sum_{k=1}^L n_k n_{k-1} = S. \quad (95)$$

Multiplying the L inequalities yields

$$d_{\text{path}} = \prod_{j=0}^{L-1} n_j \leq S^L. \quad (96)$$

Every summand in S is a positive integer, so $S \geq 1$. Since $L \leq L_0$,

$$S^{L_0} = S^L S^{L_0-L} \geq S^L. \quad (97)$$

If $L = 1$, then $n_1 = 1$ and $S = n_0 = d_{\text{path}} = S^L$. If $S = 1$, positivity forces $L = 1$ and all three quantities in (94) equal one. $\square$

Proposition A.9 (Public error-threshold specialization). *Under Assumptions 1, 2, and 4, and Proposition A.8, the same unconditional law $\mathcal{P}_{\text{gate}}$ satisfies*

$$\forall \mathcal{D} \in \Delta(\mathcal{X}) \quad \forall h \in \mathcal{H}, \quad \mathbb{E}_{\varphi \sim \mathcal{P}_{\text{gate}}} \left[\inf_{w \in \mathbb{R}^{d_{\text{path}}}} R_{\mathcal{D},h}(w, \varphi) \right] \leq 2\varepsilon \quad (98)$$

and consequently

$$\text{dc}_{2\varepsilon}(\mathcal{H}) \leq \text{dc}_{\varepsilon+\delta_0}(\mathcal{H}) \leq d_{\text{path}}. \quad (99)$$

Proof. The robust-tube range gives $0 \leq \delta_0 \leq \varepsilon$, hence

$$\varepsilon + \delta_0 \leq 2\varepsilon. \quad (100)$$

The witness bound (92) therefore proves (98) under the unchanged law and dimension. If $0 \leq a \leq b$, every law admissible at level a is admissible at level b , so

$$\text{dc}_b(\mathcal{H}) \leq \text{dc}_a(\mathcal{H}). \quad (101)$$

Apply this with $a = \varepsilon + \delta_0$ and $b = 2\varepsilon$; nonemptiness of the stricter admissible set is supplied by Proposition A.8. If $\varepsilon = 0$, then $\delta_0 = 0$ and both thresholds are zero; if $\delta_0 = \varepsilon$, they coincide. $\square$

A.8 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.8 proves (18) with one unconditional law and gives

$$\text{dc}_{\varepsilon+\delta_0}(\mathcal{H}) \leq d_{\text{path}}. \quad (102)$$

Lemma A.7 gives $d_{\text{path}} \leq S^L \leq S^{L_0}$, proving (19). Finally, Proposition A.9 reuses that same law and gives the first inequality in (20); combining it with (102) and the path-count bound gives the stated S^{L_0} corollary. All steps preserve the fixed tie rule, finite-horizon expectation mode, and initialization-first quantifier order. $\square$